Multimedia Signal Processing - 2nd Module

January, 15 2021

1. **Question (8 pts)** Consider a continuous source X that emits the following series of symbols

$$x = \{3,\ 0,\ 6,\ 6,\ 6,\ 3\}.$$

You are given a quantizer $Q(\cdot)$ that applies the following quantization rule

$$Q(x) = \left\lceil \frac{x}{\Delta} \right\rceil \cdot \Delta,$$

where $\lceil \cdot \rceil$ rounds to the least integer greater than or equal to its argument, and $\Delta = 2$ is the quantization step.

   (a) Compute the sequence $\hat{x}_{\text{PCM}}$ obtained by encoding and decoding x using a PCM scheme. Clearly specify which is the encoded and which is the decoded signal.

   (b) Compute the sequence $\hat{x}^{\text{open}}_{\text{DPCM}}$ obtained by encoding and decoding x using an open-loop DPCM scheme. The predictor acts by subtracting half of the previous sample (i.e., $\tilde{x}(n) = 0.5 \cdot x(n-1)$). Use the same quantizer defined above, and consider that the first sample is encoded using PCM. Clearly specify which is the encoded and which is the decoded signal.

   (c) Compute the sequence $\hat{x}^{\text{close}}_{\text{DPCM}}$ obtained by encoding and decoding x using a closed-loop DPCM scheme. Use the same prediction rule, quantizer and strategy for the first sample applied before. Clearly specify which is the encoded and which is the decoded signal.

   (d) Comment on the role of the first sample of the sequence for DPCM schemes. How can you typically deal with it and why?

   (e) Consider applying lossless coding to $\hat{x}_{\text{PCM}}$. Propose a prefix-code, and write the code associated to the sequence of symbols $\hat{x}_{\text{PCM}}$. Compute the bitrate in bit per symbol.

Consider the following probability mass function

$$p(0) = 0.1; \quad p(2) = 0.3; \quad p(4) = 0.4; \quad p(6) = 0.1; \quad p(8) = 0.1.$$

   (a) Using MSE as distortion metric and entropy as rate measure, draw the rate-distortion working point of each one of the three coding techniques just applied (i.e., PCM, closed-loop DPCM, open-loop DPCM).

2. **Question (7 pts)**
Oral examination.

3. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

- [T] [F] Consider a source X with alphabet $A = \{a, b, c\}$. Consider the following associated code: $C(a) = 0$, $C(b) = 10$, $C(c) = 00$. This code may have been obtained using Huffman coding scheme.
- [T] [F] Consider a source X characterized by an infinite element alphabet. It is possible to encode the output of the source with zero distortion by using lossless coding.
- [T] [F] The matrix $T = \begin{bmatrix} 1 & 1 & 0.5 \\ 2 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ cannot be used in a transform coding scheme.
- [T] [F] In a practical situation transform coding can provide more distortion than normal PCM at the same rate.
- [T] [F] It is possible to apply perceptual coding as a lossy coding technique for speech signals.
- [T] [F] The first encoded frame of a video sequence is typically an inter-coded frame.
- [T] [F] Huffman coding scheme provides the same rate of Shannon coding scheme for dyadic sources.
- [T] [F] A midtread uniform scalar quantizer has a reproduction level equal to zero.

4. **Question (Matlab - 10pts)**

Load the RGB image 'peppers.png' running the command `img = imread('peppers.png');` and follow these steps:

(a) Display the image in a figure.

(b) Compute and display the histogram of the red, green and blue channel.

(c) Compute the entropy of each channel.

(d) Convert the image to the YCbCr color space using the relation

$$\begin{bmatrix} Y \\ Cb \\ Cr \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.5 \\ 0.5 & -0.419 & -0.0813 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

(e) Considering only the luminance component, compute the 4x4 KLT matrix:

i. Extract all non-overlapping 4x4 blocks.

ii. Compute the correlation matrix.

iii. Compute the KLT matrix.

(f) Compare KLT matrix with 2D-DCT matrix of the same size in terms of MSE.

(g) Can you use the found KLT matrix to encode Cb or Cr components? What would you expect?